

TextScene: Knowledge-Enhanced Scenario Generation for Autonomous Driving via Map-Topology Abstraction and Heterogeneous Retrieval

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ABSTRACT

Large Language Models (LLMs) face three fundamental challenges in generating high-fidelity and high-value autonomous driving test scenarios: logical hallucinations due to a lack of domain risk semantics, spatial failures caused by the difficulty in comprehending complex geometric maps, and syntax errors under strict simulation standards. To address these issues, we propose TextScene, a knowledge-enhanced framework designed to transform natural language test requirements into standard-compliant OpenSCENARIO code. Unlike existing methods, our core contribution lies in the construction of three core technical components: (1) **Real-Data-Grounded Dual-Stream Retrieval-Augmented Scene Parsing**, which integrates over 8,000 real-world CIDAS accident cases reconstructed under SOTIF standards and a trigger condition library containing causal inference logic to generate structured scenes. This dual knowledge sources significantly improves long-tail coverage and testing value. (2) **Map-Topology Abstraction Algorithm**, which compresses complex OpenDRIVE parameters into semantic topological graphs. This endows LLMs with explicit spatial reasoning capabilities to substantially reduce spatial hallucinations. (3) **Syntax-Constrained Instruction-Tuning Paradigm**, which is built upon strictly verified "physical scene-standard code" pairs. Through Parameter-Efficient Fine-Tuning (PEFT), this paradigm enables the model to learn the complex nested syntax of OpenSCENARIO. Furthermore, TextScene introduces a Heterogeneous Retrieval-Augmented Generation (RAG) mechanism to achieve physical grounding across empirical accident data, risk mechanisms, and map topological constraints. Extensive experiments demonstrate that TextScene achieves a Compilation Rate (CR) of 90.6% and a Simulation Success Rate (SSR) of 87.9%, significantly outperforming GPT-5 and existing retrieval-based baselines in terms of physical validity and logical fidelity.

1. Introduction

The safety validation of Autonomous Driving Systems (ADS) fundamentally depends on massive and diverse testing scenarios (Aasi et al., 2025; Guo et al., 2024; Ma et al., 2024). Given the high economic costs and potential safety risks of real-world road testing, virtual testing based on simulation engines (such as CARLA (Dosovitskiy et al., 2017)) has become the established standard paradigm in the industry (Zhang et al., 2024; Ding et al., 2024; Wei et al., 2024). In recent years, Generative AI (GenAI), especially LLMs, has demonstrated immense potential in translating natural language requirements into simulation scenarios (Yang et al., 2024; Zhang et al., 2026a; Sheng et al., 2026), leveraging their exceptional semantic understanding and generative capabilities. This "Text-to-Scene" paradigm promises to significantly lower the construction barrier for long-tail test cases. However, directly utilizing general-purpose LLMs to generate professional-grade simulation scenarios faces a substantial semantic-physical gap (Zhang et al., 2024; Cai et al., 2025; Zhang et al., 2026b). As shown in Figure 1 (left side), existing methods are primarily constrained by three core dilemmas:

- **Lack of Realism and Risk:** Due to the scarcity of domain-specific risk semantics, general models tend to generate simple scenarios characterized by simple scenarios with weak causal structure and limited interaction complexity, missing the critical game-theoretic and adversarial interactions found in real-world accidents, resulting in low-value generated test cases.

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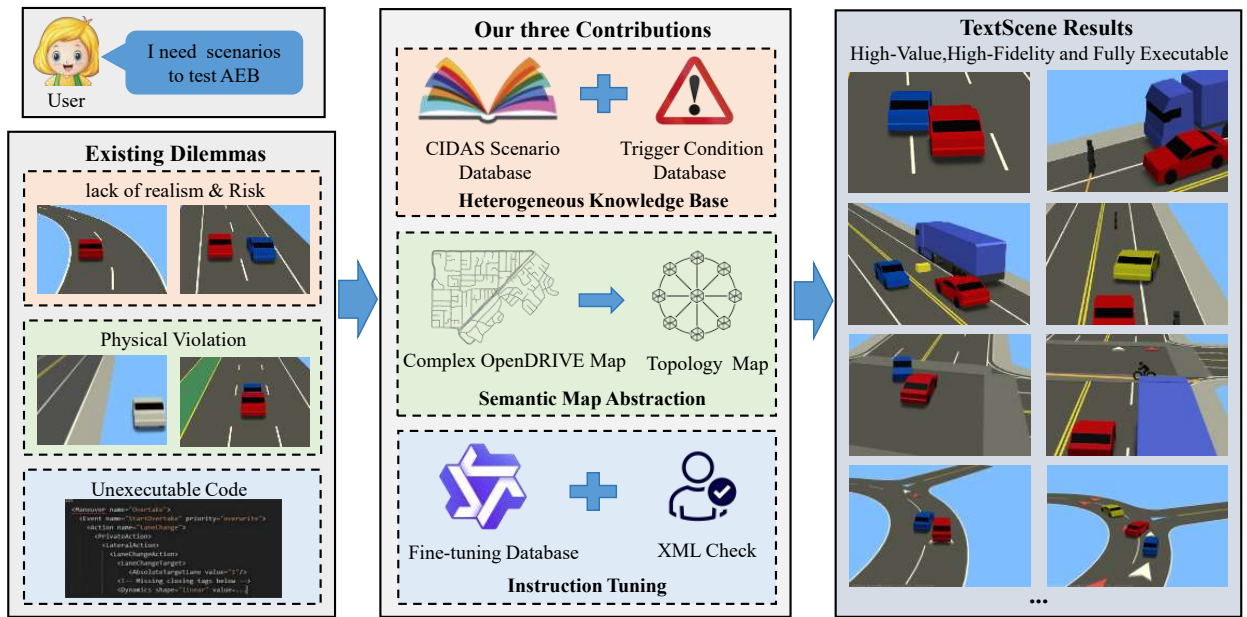


Fig. 1. The TextScene framework addresses existing dilemmas in scenario generation (Left) by leveraging three core contributions (Middle) to produce high-value, high-fidelity, and highly executable simulation scenarios (Right).

- **Violation of Physical Laws:** LLMs lack an understanding of the complex geometric space of maps and frequently suffer from spatial hallucinations, causing generated vehicles to exhibit severe violations of physical common sense, such as overlapping, retrograde driving, or drifting off the roadway.
- **Unexecutable Code:** Industrial-grade simulation standards (such as OpenSCENARIO) possess complex nested structures and strict syntactic constraints. Code generated by general-purpose models often cannot be correctly parsed by simulators due to tag closure errors or missing key attributes.

To address these challenges, we introduce TextScene, a knowledge-enhanced framework designed to translate natural language requirements into industrial-grade simulation code with improved reliability, as shown in Figure 1 (middle). The core innovation of TextScene lies in the construction of three key data infrastructures. First, we propose Real-Data-Grounded Dual-Stream Retrieval-Augmented Scene Parsing which integrates CIDAS real-world accident data with risk-triggering conditions, thereby providing the model with references for "high-value" critical scenarios and causal chains of risk logic. Second, we introduce Map-Topology Abstraction Algorithm to compress complex OpenDRIVE specifications into semantic topological graphs, endowing the model with spatial reasoning capabilities. Finally, we propose a Syntax-Constrained Instruction-Tuning Paradigm, which leverages a constructed "physical scenario-standard code" alignment dataset alongside a rigorous OpenSCENARIO XML syntax validation mechanism to enable the model to master standard OpenSCENARIO syntax rules. By bridging the semantic-physical gap, TextScene translates abstract natural language requirements into standard-compliant OpenSCENARIO code, achieving a Compilation Rate of 90.6% and a Simulation Success Rate of 87.9% on the 1,000 evaluation scenarios. As shown in Figure 1 (right side), compared to existing baseline methods, scenarios generated by TextScene not only achieve high executability but also demonstrate high-fidelity physical interactions and high-value testing content (e.g., multi-vehicle dynamic interactions under adverse weather conditions). In summary, the main contributions of this paper are as follows:

- **Real-Data-Grounded Dual-Stream Retrieval-Augmented Scene Parsing:** Integrating over 8,000 SOTIF-compliant CIDAS cases and a causal trigger condition library, this mechanism ensures physical authenticity and enhances long-tail coverage by injecting adversarial risk elements.
- **Map-Topology Abstraction Algorithm:** We propose a Geometry-to-Topology algorithm that compresses OpenDRIVE data into semantic graphs. This empowers LLMs with spatial reasoning, substantially reducing hallucinations such as vehicle overlaps and off-road driving.

- **Syntax-Constrained Instruction-Tuning Paradigm:** Leveraging QLoRA on a validated "physical scenario-standard code" dataset, this paradigm improves the model's ability to generate OpenSCENARIO's complex nested structures, ensuring both syntactic correctness and code executability.

2. Related Works

2.1. Traditional Safety-Critical Scenario Generation

The generation of safety-critical scenarios constitutes a fundamental pillar for validating ADS. Traditional methodologies are predominantly classified into three paradigms. Data-driven approaches leverage generative models, such as Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs), to synthesize traffic flows derived from real-world datasets like Waymo or NuScenes (Yang et al., 2020; Ding et al., 2020; Scanlon et al., 2021; Caesar et al., 2020). Although effective at capturing realistic data distributions, these methods exhibit limitations in generating long-tail scenarios, primarily due to the inherent sparsity of long-tail scenarios in accident data (<20% coverage) (Cai et al., 2025). Adversarial generation frames scenario creation as an optimization problem, seeking trajectories that minimize the ego vehicle's safety metrics (Abeysirigoonawardena et al., 2019; Lee et al., 2015; Prakash et al., 2019; Zhang et al., 2023). While methods like STRIVE (Rempe et al., 2022) optimize latent spaces to induce collisions, they often yield adversarial attacks that, despite being physically feasible, lack behavioral realism. Knowledge-based methods construct scenarios based on explicit ontologies and traffic regulations (Bagschik et al., 2018; Cai et al., 2020; Rocklage et al., 2017). For instance, (Klischat et al., 2020) extracts road topologies from OpenStreetMap. While these rule-based templates ensure logical consistency, they typically lack the stochastic diversity and complexity necessary to simulate unpredictable, human-like violations. In contrast, TextScene aims to preserve the empirical realism of accident data while enabling semantic and compositional generation of low-frequency safety-critical scenarios through causal trigger grounding.

2.2. LLM-Driven Scenario Synthesis and Code Generation

The proliferation of LLMs has transitioned the field from optimization methodologies to a semantic generation paradigm (Hou et al., 2025; Kuang et al., 2025). Initial explorations, including ChatScene (Zhang et al., 2024) and Text2Scenario (Cai et al., 2025), utilize LLMs to translate textual requirements into intermediate Domain Specific Languages (DSLs). Specifically, ChatScene leverages the general knowledge of LLMs to generate natural-language descriptions of safety-critical scenarios from user instructions, and retrieves and assembles corresponding Scenic code fragments from a pre-constructed database into executable scripts. Talk2Traffic (Sheng et al., 2025) uses a specialized interpreter to extract structured traffic-scene representations from multimodal inputs, which are then converted into executable Scenic code through a RAG-based retrieval and generation process. LeGEND (Tang et al., 2024) further refines this by employing a dual-LLM architecture to decouple logical reasoning from parameter mapping. More recently, research focus has shifted toward maximizing scenario criticality: Gao et al. (Gao et al., 2025) encode collision constraints via Chain-of-Thought (CoT) (Wei et al., 2022) prompting to steer adversarial generation, whereas RealGen (Ding et al., 2024) and AGENTS-LLM (Tan et al., 2023) investigate retrieval-based context learning. However, a critical disconnect persists between high-level semantics and low-level execution. Predominant methods largely rely on simplified DSLs or generic map templates, thereby bypassing the intricate geometric and syntactic constraints of industry standards (OpenDRIVE/OpenSCENARIO) (ASAM e.V., 2020). Without explicit grounding in complex map topologies and strict syntax verification, such methods are prone to *spatial hallucinations* and *compilation failures*, rendering them unsuitable for high-fidelity industrial simulation. TextScene differs from these LLM-driven approaches by coupling semantic generation with map-topology alignment and syntax-constrained OpenSCENARIO generation, rather than relying on domain-specific language (DSL) code snippet retrieval and simple combination.

2.3. Knowledge Retrieval and Map Grounding for Scenario Generation

Retrieval-Augmented Generation (RAG) has proven effective in grounding LLMs within domain-specific contexts. In the ADS domain, recent efforts have integrated knowledge graphs or regulatory rulebooks to mitigate hallucinations (Mei et al., 2025; Chen et al., 2024; Míguelañez et al., 2011). However, a critical bottleneck persists in Spatial Map Reasoning. Industry-standard High-Definition (HD) maps consist of extensive geometric definitions, such as clothoids and polynomials, that often exceed the context window constraints of LLMs, making direct processing infeasible. Existing approaches typically compromise by either encoding maps as rasterized images (Li, 2024) (Xia et al., 2026) (sacrificing topological precision) or bypassing map constraints entirely via unstructured open spaces (Chang et al.,

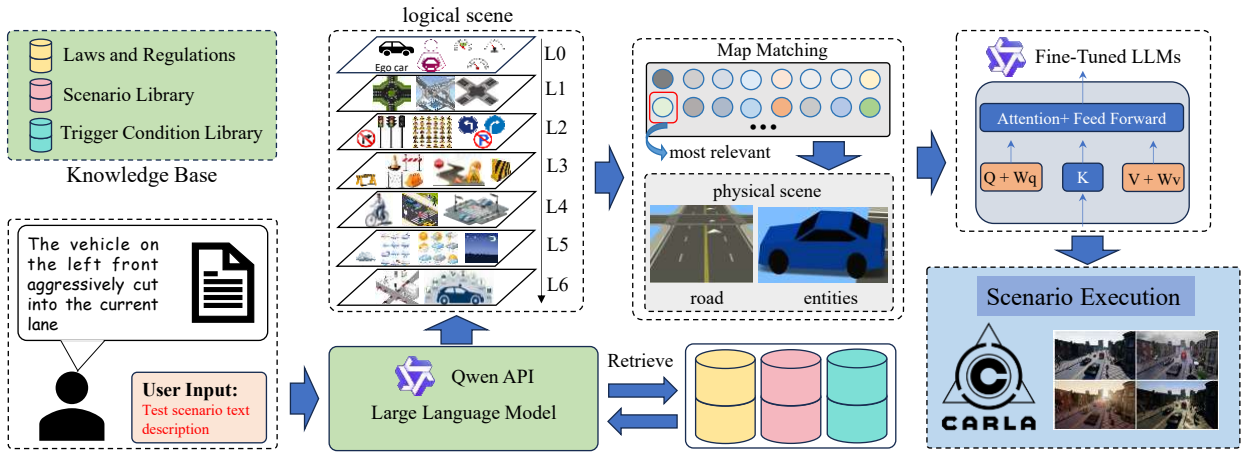


Fig. 2. The overall framework of TextScene. It consists of three progressive stages: (1) Retrieving real-world knowledge to construct a logical scenario; (2) Aligning with map topology to instantiate physical parameters; (3) Generating standard OpenSCENARIO code via a fine-tuned LLM.

2024). TextScene addresses this deficiency. Diverging from prior works, we introduce a semantic abstraction layer that transforms intricate map geometries into structured topological graphs. This empowers the LLM to perform explicit spatial reasoning, thereby improving spatial validity and executable OpenSCENARIO generation.

Table 1

The L0-L6 hierarchical scenario structure based on the SOTIF standard

Layer	Name	Description and Examples
L0	Internal ego-vehicle	Ego-vehicle kinematics, system status, and driver behavior.
L1	Road-level	Static road geometry, topology, surface characteristics, and lane definitions.
L2	Traffic infrastructure	Static roadside infrastructure and associated traffic rules.
L3	Temporary manipulation	Temporary modifications to L1/L2, such as roadworks, construction zones.
L4	Objects	Dynamic entities (e.g., vehicles, pedestrians) and their interacting trajectories.
L5	Environment	Weather, lighting, visibility, temperature, and time of day.
L6	Communication	V2X information, digital map.

3. Methodology

In this study, we introduce **TextScene**, a knowledge-enhanced framework intended to facilitate the translation of unstructured natural language test requirements into standardized simulation codes, thereby bridging the inherent semantic gap. As illustrated in Figure 2, the framework employs a *hierarchical progressive generation paradigm* to instantiate user instructions as executable simulation scenarios. To avoid terminological ambiguity, we clarify the scenario concepts used in this paper. Safety-critical scenarios refer to scenarios in which the autonomous driving system faces potential safety threats, such as possible collisions or failures to respond appropriately to surrounding road users. Long-tail scenarios refer to scenarios that occur infrequently in naturalistic driving data but exhibit high diversity and complexity. These two concepts characterize scenarios from different perspectives: the former emphasizes safety relevance, whereas the latter emphasizes rarity in the data distribution.

3.1. Real-World Accident Scenario Knowledge Base

To mitigate LLM hallucinations during scenario generation, we constructed a high-fidelity, long-tail knowledge base derived from real-world accidents. We detail this knowledge base through three key dimensions: ontology-based knowledge representation, the construction methodology, and a diversity analysis of the scenario data distribution.

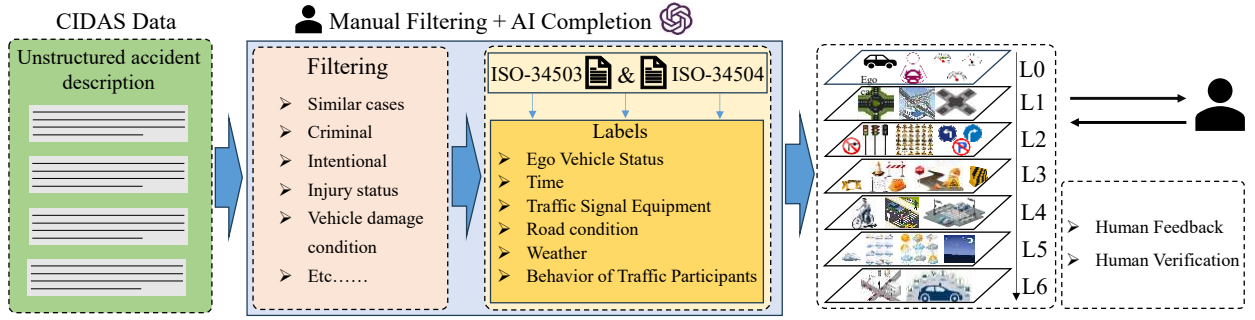


Fig. 3. The Human-in-the-loop construction pipeline. It transforms unstructured CIDAS data into structured JSON knowledge through filtering, ISO-standard labeling, and AI-assisted completion.



Fig. 4. The Cross-Modal Knowledge Extraction Pipeline. It utilizes Vision-Language Models (VLMs) to extract causal triplets (Source, Mechanism, Consequences) from heterogeneous visual data sources.

3.1.1. Ontology-based Knowledge Representation

To resolve the ambiguity of unstructured text, we adopt the SOTIF ([International Organization for Standardization, 2022](#)) L0-L6 layered model as our foundational ontology. As shown in Table 1, this framework decomposes scenarios into seven orthogonal layers, ranging from ego-vehicle kinematics (L0) and static road infrastructure (L1-L3) to dynamic interactions (L4) and environmental/digital information (L5-L6). This *semantic disentanglement* creates a standardized schema of "Feature Slots," compelling the LLM to define all essential simulation parameters and preventing physical inconsistencies caused by information omission.

3.1.2. Human-in-the-loop Construction Pipeline

The raw data for this study is derived from the CIDAS (Chinese Traffic Accident Deep Study) database. To convert this raw corpus into structured knowledge, we developed a "Human-in-the-loop" (HITL) hybrid processing pipeline, as illustrated in Figure 3. This pipeline comprises three primary phases:

- **Heuristic Filtering:** We apply a set of exclusion criteria to discard invalid samples, such as those involving criminal activities, ambiguous intentions, thereby isolating cases with significant testing value.
- **Standardized Mapping:** We adopt the ISO 34504 (Scenario Categorization) ([International Organization for Standardization, 2023](#)) as the foundational annotation schema to unify the terminology used within the accident descriptions.
- **Semantic Completion:** To address implicit information frequently omitted in raw reports, we leverage LLMs to infer missing contextual details. If the accident data already contain L0-L6 hierarchical structure information, the original data are retained without modification; for missing L0-L6 information, the LLM is used to infer and complete the missing content. Following manual verification, each case is instantiated into a standardized L0-L6 JSON format. This procedure guarantees both data *completeness* and *consistency*.

3.1.3. Diversity and Long-tail Case Analysis

As shown in Figure 5 (a), a statistical analysis of over 8,000 structured cases reveals a pronounced long-tail distribution. The repository is enriched with 26.4% high-risk interactions and 24.1% Vulnerable Road User (VRU)

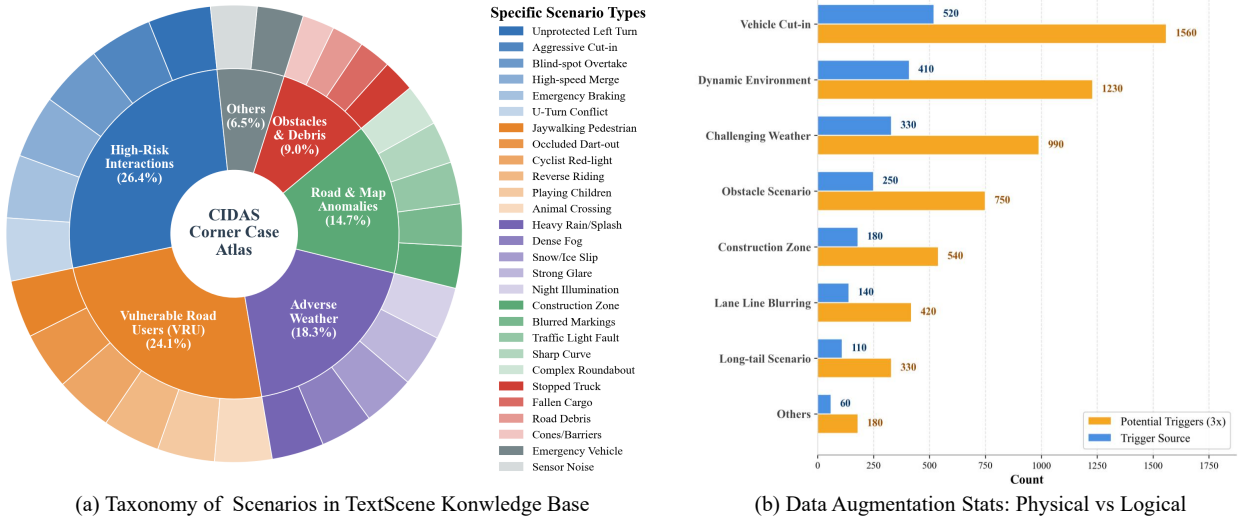


Fig. 5. Taxonomy of long-tail scenarios and safety-critical scenarios in the scenario knowledge base. The distribution shows a strong skew towards high-risk and long-tail scenarios.

scenarios. Furthermore, adverse weather conditions and road anomalies collectively constitute 33% of the scenario library. This highly diverse and long-tail skewed distribution provides a repository of challenging, high-quality prior knowledge for the subsequent RAG process, ensuring that the generated scenarios effectively validate the robustness of autonomous driving algorithms under complex boundary conditions.

3.2. Risk-Aware Trigger Condition Knowledge Base

To enhance testing criticality, we superimpose risk factors onto routine maneuvers. We construct a trigger condition knowledge base encoded as "Source-Mechanism-Consequence" causal chains via cross-modal extraction and logical augmentation.

3.2.1. Cross-Modal Knowledge Extraction Pipeline

We extract implicit risk knowledge from unstructured visual data using a Visual-Language Co-Extraction Pipeline, as illustrated in Figure 4.

The raw data originates from two distinct sources: a proprietary real-world database capturing diverse long-tail scenarios, and established open-source datasets such as nuScenes and Waymo. This corpus encompasses representative perception and control challenges, such as aggressive maneuvers or adverse weather conditions. Subsequently, semantic parsing is performed using State-Of-The-Art (SOTA) Visual Language Models (VLMs) as feature extractors (Bai et al., 2025) to transmute continuous video sequences into discrete natural language descriptions. The VLM is prompted to identify scene-specific risk factors and generate preliminary textual annotations. These unstructured descriptions are standardized into causal triplets formatted as <Source, Mechanism, Consequence>. For instance, an illegally parked truck is categorized as the "Source", the resulting visual occlusion is defined as the "Mechanism", and the potential pedestrian intrusion is designated as the "Consequence". This structured representation serves as a foundational reference for the LLM to generate high-risk scenarios.

3.2.2. Expansion of Potential Trigger Condition

To address the disparity between the finite nature of physical data collection and the unbounded variability of real-world risks, we propose a logical reasoning mechanism grounded in the constructed causal triplets to achieve significant data augmentation. across participant roles, relative speeds, cut-in intervals, occlusion conditions, adverse weather, and response delays under traffic-feasibility constraints. As illustrated in Figure 5 (b), by establishing a mapping from the physical to the logical domain, each physical trigger source yields, on average, more than a three-fold increase in potential logical variations, with invalid or physically implausible combinations filtered before storage. For instances, in the "Vehicle Cut-in" category, 520 physical samples were logically extrapolated into 1,560 complex combinations, encompassing diverse relative speeds, cut-in intervals, and subsequent maneuvers. This "analogy-based" reasoning

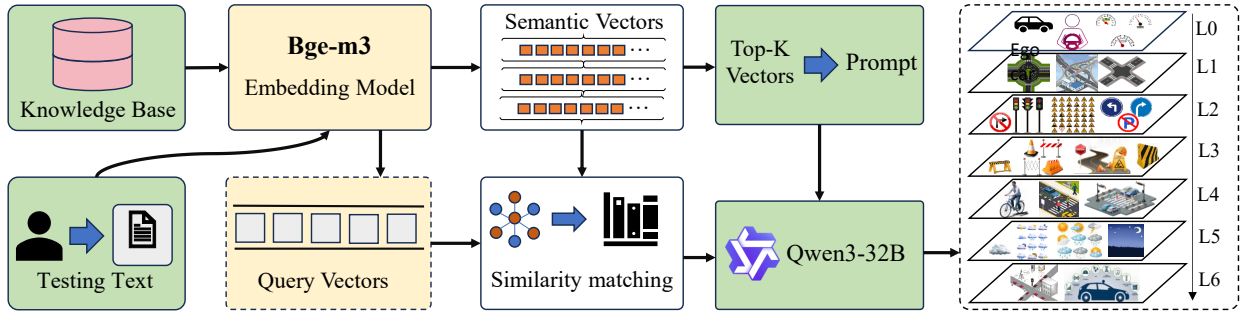


Fig. 6. The knowledge-retrieved logical scenario generation process. It utilizes embedding models to map knowledge and queries into a semantic space, employing a dual-stream retrieval strategy to prompt the LLM.

capability effectively broadens the search space for scene generation, enabling the system to synthesize complex adversarial scenarios that transcend the coverage of the original training data.

3.3. Knowledge-Retrieved Logical Scenario Generation

To mitigate domain knowledge deficits and hallucinations in general LLMs, we propose a RAG mechanism for scenario generation (Lewis et al., 2020). Acting as *semantic middleware*, it maps abstract requirements into L0-L6 logical scenarios, ensuring both statistical authenticity and physical completeness, as shown in Figure 6.

The pipeline serializes the descriptions of all scenarios in the Real-World Accident Scenario Knowledge Base and the descriptions of all trigger conditions in the trigger-condition library, mapping them to dense vectors via a pre-trained embedding model. In parallel, user instructions are encoded into query vectors, enabling the retrieval of the most semantically aligned real scenarios and trigger conditions via cosine similarity. The retrieval process can be expressed as:

$$index = \text{TopK}\left(\left\{\frac{e_{b,i}^T e_q}{\|e_{b,i}\| \cdot \|e_q\|}\right\}_{i=1}^N\right) \quad (1)$$

where *index* represents the indices of the retrieved scenarios or trigger conditions, $e_{b,i}$ represents vector data from the knowledge base, e_q represents the query vector input by the user.

To reconcile “physical fidelity” with “testing criticality” during generation, we devised a dual-stream retrieval strategy. This mechanism executes two parallel queries:

- **Scenario Retrieval:** This process extracts the Top-*K* real-world accident cases from the CIDAS library to establish foundational spatiotemporal features and game-theoretic logic.
- **Trigger Condition Retrieval:** This mechanism identifies the top-*K* relevant risk factors to introduce potential testing challenges.

These contexts act as “knowledge anchors,” ensuring the realism of the generated scenarios and the incorporation of risk factors. The retrieved top-*K* real-world scenario data and trigger conditions are formatted into a prompt template to establish an in-context learning (ICL) paradigm. By integrating these references with the L0-L6 JSON constraints, the large language model can synthesize user requirements and empirical accident logic to produce a structured logical scenario.

3.4. Map-Topology Alignment and Physical Instantiation

Standard OpenDRIVE files contain extensive geometric definitions with low semantic density, making direct ingestion prone to context overflow and poor interpretability. To address this, we propose Map-Topology Abstraction algorithm that transforms large-scale networks into a lightweight topological knowledge base tailored for LLM reasoning, as shown in Figure 7.

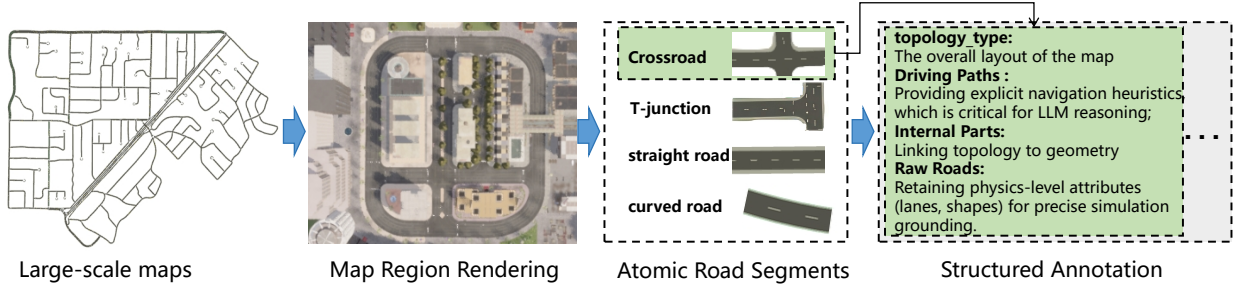


Fig. 7. The semantic abstraction of HD Maps. The process segments large-scale maps into atomic units and extracts structured topological annotations to facilitate LLM reasoning.

3.4.1. Atomic Road Segmentation

To simplify spatial reasoning, we decompose the road network using topological junction nodes as boundaries, partitioning it into discrete *Atomic Road Segments*. These units are categorized into primitives (e.g., crossroads, T-junctions) as shown in Figure 7. This atomization transforms global road matching into a localized task, significantly reducing LLM reasoning overhead.

3.4.2. Topological Graph Extraction and Annotation

To bridge the gap between OpenDRIVE geometry and LLM reasoning, we reformulate map data into a three-level hierarchical JSON schema, extracting core spatial information, as shown in Figure 7.

- **Scenario Summary:** Provides macroscopic statistics on map typologies and scale, equipping the LLM with a global prior for rapid environment identification.
- **Intersection Semantics:** Bridging macro- and micro-levels, this layer abstracts branching structures into explicit traversal paths (e.g., From $Road_2$ to $Road_1$). It replaces implicit geometry with navigational heuristics, enabling direct high-level semantic queries and bypassing complex parsing.
- **Raw Roads Physicality:** Detailing physical attributes and connectivity, this layer retains geometric fidelity (shape, length, and lane configuration) and constructs a directed graph vialink annotations. This ensures physical feasibility and provides precise spatial references for localization and planning.

This abstraction achieves a compression rate exceeding 90% while preserving connectivity and interpretability, establishing an efficient, LLM-comprehensible, and retrieval-friendly semantic map knowledge base.

3.4.3. Topology-Aware Physical Scenario Instantiation

Leveraging the semantic map library, we implement a topology-aware framework synergizing semantic matching with spatial reasoning. During retrieval, the system identifies atomic road segments based on logical scene descriptions. Subsequently, via annotation-driven reasoning, the LLM exploits pre-annotated traversal paths to ground abstract entities onto specific Lane IDs (e.g., mapping target vehicle to $Road_1$, $Lane_2$). Integrating this context, the LLM infers relative positions and behaviors, instantiating physical parameters and orchestrating event triggers. This effectively bridges the semantic-geometric divide, transforming abstract language into parameterized physical scenarios.

3.5. Syntax-Constrained Code Generation via Instruction Tuning

Translating physical parameters into executable code remains challenging. OpenSCENARIO's intricate XML nesting and strict constraints often overwhelm general LLMs, causing syntactic failures like unclosed tags or parameter hallucinations. To bridge this "scene-to-code" gap, we propose a QLoRA (Dettmers et al., 2023) based instruction fine-tuning framework, as illustrated in Figure 8.

3.5.1. Dataset Construction

To ensure the model adheres to complex syntactic rules, we constructed a high-quality instruction fine-tuning dataset comprising "physical scene–scene code" pairs. Specifically, the input data is derived from the structured JSON

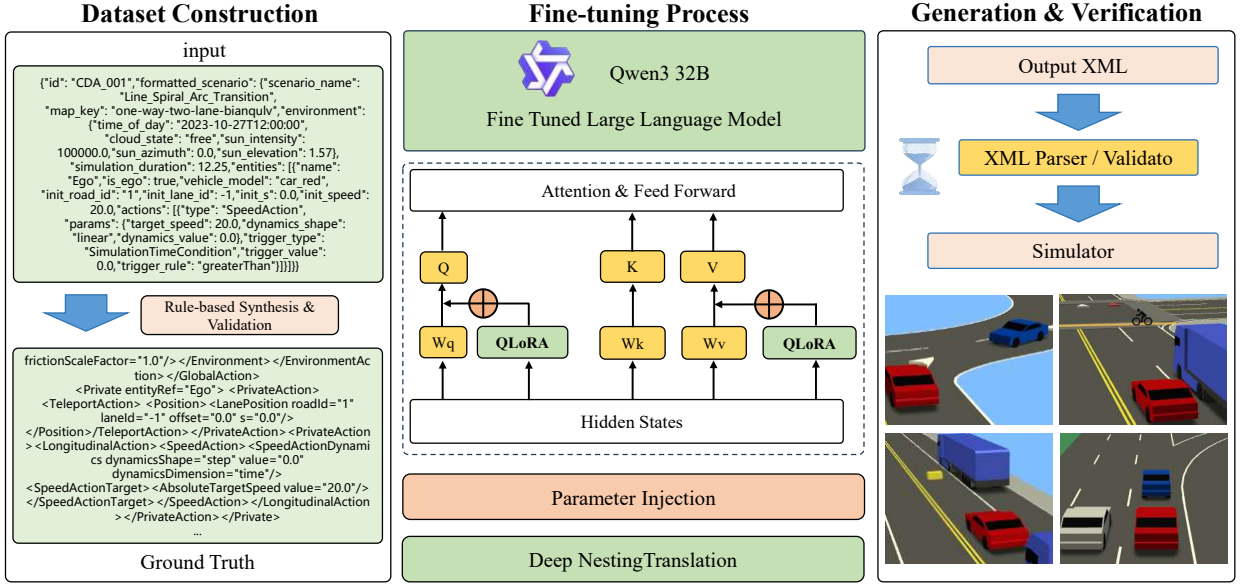


Fig. 8. The instruction tuning framework. It bridges the syntax gap by fine-tuning the LLM on high-quality synthetic datasets and verifying the output through a closed-loop system.

physical scenes generated in Section 3, as shown in Figure 8. The corresponding OpenSCENARIO output is produced through a hybrid approach combining LLM generation with manual refinement. Crucially, all OpenSCENARIO XML scripts underwent rigorous validation via simulation engines and expert inspection to guarantee zero syntactic errors and full logical correctness.

3.5.2. QLoRA Fine-tuning for Code Generation

We fine-tune Qwen3-32B (Yang et al., 2025) using QLoRA —an efficient extension of LoRA (Hu et al., 2022). The process of LoRA can be expressed as:

$$y = W_0x + \Delta Wx = W_0x + BAx \quad (2)$$

where $W_0 \in \mathbb{R}^{d \times k}$ represents the pre-trained model weights, $\Delta W \in \mathbb{R}^{d \times k}$ represents the weight update value, $B \in \mathbb{R}^{d \times r}$ and $A \in \mathbb{R}^{r \times k}$ represent the introduced low-rank decomposition matrices.

QLoRA employs 4-bit NormalFloat (NF4) quantization, which is theoretically optimal for normally distributed weights, alongside a double quantization mechanism for the quantization constants. Specifically, we integrated low-rank adapters ($r = 16$) into the query, key, and value projections of each transformer layer. The formulation of the QLoRA process is defined as follows:

$$W^{\text{BF16}} = \text{doubleDequant}(c_1^{\text{FP32}}, c_2^{\text{FP8}}, W^{\text{NF4}}) \quad (3)$$

$$Y^{\text{BF16}} = X^{\text{BF16}} W^{\text{BF16}} + X^{\text{BF16}} L_1^{\text{BF16}} L_2^{\text{BF16}} \quad (4)$$

where c_1^{FP32} represents the quantization constant for the second quantization, c_2^{FP8} represents the quantization constant obtained after the second quantization, W^{NF4} represents the pre-trained model weights parameters quantized to 4-bit NormalFloat, and L_1^{BF16} and L_2^{BF16} respectively represent the introduced low-rank decomposition matrices.

4. Experiments

This chapter presents a comprehensive validation of the proposed TextScene framework. The experimental evaluation is explicitly designed to address the following three core Research Questions (RQs):

Table 2

The selected 20 test functional tasks

Category	Test Task Name	Risk Level
Highway & Basic Maneuvers	Automated Lane Keeping	Low
	Adaptive Cruise Control	Medium
	Proactive Lane Change	Medium
	High-speed Overtaking	High
	Highway On-Ramp Merging	High
	Highway Off-Ramp Exiting	Medium
	Traffic Jam Pilot	Medium
Intersection & Junctions	Signalized Intersection Navigation	Medium
	Unsignalized Intersection Navigation	High
	Roundabout Negotiation	High
	Unprotected Left Turn Decision	High
	U-Turn at Intersection	High
Dynamic Interactions	Response to Vehicle Cut-in	High
	Response to Lead Vehicle Cut-out	High
	Vulnerable Road User Interaction	Medium
	Emergency Hazard Avoidance	High
	Response to Special Vehicles	Low
Static & Environmental	Construction Zone Navigation	Medium
	Narrow Road Negotiation	Medium
	Tunnel Passage	High

- **RQ1:** How does the proposed framework compare against state-of-the-art (SOTA) LLMs and existing scene generation baselines in terms of the executability and fidelity of the generated scenarios?
- **RQ2:** What is the distinct impact of the framework’s core components—specifically, RAG-based knowledge retrieval, Map-Topology Abstraction, and instruction fine-tuning—on the overall system performance?
- **RQ3:** To what extent does the framework demonstrate robustness across varying hyperparameter configurations and diverse map topological complexities?

4.1. Experimental Setup

4.1.1. Datasets

The experimental evaluation relies on the three core datasets established in Section 3: the **Scene Knowledge Base**, comprising 8,240 curated real-world accident cases; the **Trigger Condition Library**, containing 1,560 logical trigger variants; and the **Instruction Fine-tuning Dataset**, consisting of 8,000 high-fidelity <physical scene, OpenSCENARIO> pairs.

We randomly partitioned the fine-tuning dataset into training (6,400), validation (800), and testing (800) subsets following an 8:1:1 ratio. Furthermore, we selected the five largest standard CARLA OpenDRIVE maps and partitioned them by intersection topology into 50 map segments. These maps encompass a broad spectrum of topological structures, including highways, urban intersections, and roundabouts, etc.

4.1.2. Evaluation Metrics

To assess executability, we define **Compilation Rate (CR)** as the proportion of the 1,000 scenario-map evaluation pairs whose generated .xosc files pass XML schema verification, and **Simulation Success Rate (SSR)** as the proportion of these pairs that execute in CARLA (Dosovitskiy et al., 2017) without physical violations, such as initialization collisions or map misalignments. For fidelity, **Semantic Consistency (SC)** employs GPT-5.1 (Singh et al., 2025) to score prompt alignment on a 1-5 scale via the "LLM-as-a-Judge" paradigm, evaluated on the 1000 generated scenarios. Finally, testing value is evaluated via **Mean Minimum TTC (m-TTC)**, computed by averaging the minimum TTC observed within each executed scenario (lower values indicate higher risk).

To quantify the diversity of scenarios generated for the same requirement, we introduce Intra-List Diversity (ILD). For M scenarios generated under an identical requirement, we compute the maximum cosine similarity to any other in the set as the self-similarity.

$$sim_{sce,j} = \max_{j' \in \{1, \dots, M\} \setminus j} \frac{e_{sce,j'}^T e_{sce,j}}{\|e_{sce,j'}\| \cdot \|e_{sce,j}\|} \quad (5)$$

where $sim_{sce,j}$ represents the similarity score between the embedding of a certain scenario $e_{sce,j}$ and the embedding of the other generated scenarios $e_{sce,j'}$. The overall self-similarity is the mean of these values:

$$sim_{sce} = \text{mean}([sim_{sce,1}, \dots, sim_{sce,M}]) \quad (6)$$

where sim_{sce} represents the self-similarity among the generated M scenarios. ILD is defined as $1 - sim_{sce}$. Higher ILD reflects richer variation while preserving criticality—a key advantage over prior prompt-only approaches.

4.1.3. Implementation Details

In this study, we employed **bge-m3** (Chen et al., 2025) as the embedding model and **Qwen3-32B** (Yang et al., 2025) as the backbone LLM. To facilitate the translation from physical scenes to executable code, we fine-tuned Qwen3-32B utilizing the QLoRA technique. The base model was loaded with 4-bit NormalFloat (NF4) quantization. We configured the low-rank adapter with a rank of 16 and a scaling factor of 32. Training was optimized using *pagedadamw32bit* with a learning rate of 1e-4 and a batch size of 2 over 5 epochs. For generation, we fixed the temperature at 0.6 based on preliminary trials: higher values produced overly divergent and unstable scenarios, whereas lower values led to conservative and repetitive outputs. All experiments were conducted on a cluster of four RTX 3090 GPUs.

Table 3

Performance comparison with state-of-the-art baselines. **CR**: Compilation Rate; **SSR**: Simulation Success Rate; **SC**: Semantic Consistency; **m-TTC**: Mean Minimum Time-To-Collision (Lower indicates higher risk); **ILD**: Intra-List Diversity.

Method	Backbone	Setting	Executability		Fidelity	Testing Value	
			CR (%)↑	SSR (%)↑	SC (1-5)↑	m-TTC (s)↓	ILD↑
<i>SOTA General LLMs</i>							
GPT-5.1 (Singh et al., 2025)	GPT-5.1	Few-shot	58.2	34.6	4.65	5.12	0.53
Claude Opus 4.5 (Anthropic, 2025)	Claude-3	Few-shot	56.7	32.1	4.48	5.28	0.51
Gemini 3.0 Pro (DeepMind, 2025)	Gemini-3	Few-shot	71.5	48.9	4.78	4.05	0.63
Gemini 3.0 Flash (DeepMind, 2025)	Gemini-3	Few-shot	54.3	29.8	4.31	5.41	0.48
Grok 4.1 (xAI, 2025)	Grok-4	Few-shot	55.9	31.4	4.62	5.09	0.51
DeepSeek V3.2 (Liu et al., 2024)	DeepSeek-3	Few-shot	57.4	35.2	4.55	4.87	0.54
Qwen 3 Max (Yang et al., 2025)	Qwen-3	Few-shot	59.8	37.6	4.59	4.62	0.57
<i>Domain-Specific Methods</i>							
Text2Scenario (Cai et al., 2025)	Llama-3-70B	Few-shot + CoT	62.1	57.8	4.19	4.23	0.42
ChatScene (Zhang et al., 2024)	GPT-4-Turbo	Few-shot + CoT	64.7	61.3	4.32	3.68	0.45
TextScene (Ours)	Qwen3-32B	CoT+Fine-tune	90.6	87.9	4.51	2.31	0.56

4.2. Performance Analysis

We established a benchmark suite of 20 SOTIF-derived functional tasks to evaluate TextScene, as shown in Table 2. For quantitative evaluation, we used 20 functional tasks, each with 10 paraphrased prompts, yielding 200 prompt-level generated scenarios. Each prompt instance was matched to five compatible map segments drawn from the 50-segment pool, yielding 1,000 scenario-map evaluation pairs in total. The evaluation encompasses two dimensions: comparative benchmarking against SOTA baselines for executability and logical consistency, and intrinsic analysis of testing utility regarding risk coverage and diversity.

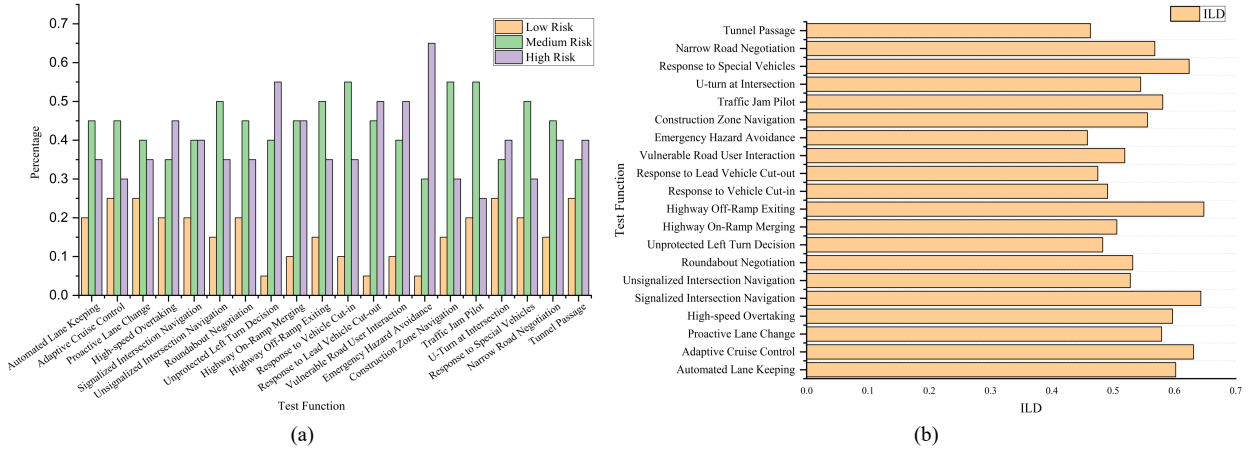


Fig. 9. Risk and diversity analysis of generated scenarios corresponding to the selected 20 functional test requirements. Subfigure (a) shows the risk level analysis results of the generated scenarios; Subfigure (b) shows the diversity analysis results of the generated scenarios.

4.2.1. Compare with State of The Art

We benchmark TextScene against two baseline categories: (1) General-purpose LLMs (GPT-5.1, Claude Opus 4.5, Gemini 3.0 Pro) utilizing Few-shot prompting with map segments obtained by our method, and (2) Domain-specific methods (Text2Scenario, ChatScene) keep its origin method utilizing Few-shot + CoT prompting. Table 3 summarizes the comparative results, focusing on code executability and physical fidelity.

Analysis of Executability: As shown in Table 3, TextScene significantly outperforms all baselines. General LLMs struggle with the rigorous XML syntax of OpenSCENARIO, with even the top-performing Gemini 3.0 Pro achieving only a 71.5% CR. In contrast, TextScene attains a CR of 90.6% and an SSR of 87.9% on the 1,000 scenario-map benchmark evaluation pairs, validating the efficacy of our instruction tuning in aligning outputs with ASAM standards. Furthermore, our method surpasses the leading domain-specific baseline, ChatScene, by 26.6 percentage points in SSR. This advantage stems from the Map-Topology Abstraction module, which substantially reduces coordinate hallucinations and off-road violations common in baselines lacking geometric grounding.

Analysis of Logical Consistency: With respect to Semantic Consistency, Gemini 3.0 Pro achieves the highest SC score of 4.78 under few-shot prompting without any domain-specific adaptation. Our fine-tuned Qwen3-32B model attains an SC of 4.51. However, that this comparison is not directly analogous and should not be construed as evidence of competitive parity. Rather, the result indicates that domain-specific instruction tuning via QLoRA preserves the model's semantic fidelity to a large extent, which suggests that catastrophic forgetting is effectively mitigated. The negligible gap of 0.08 relative to the untuned Qwen3-Max baseline provides further corroborating evidence for this interpretation.

Analysis of Testing Value: As shown in Table 3, TextScene achieves the lowest m-TTC of 2.31 s among all evaluated methods, representing a 37.2% reduction relative to the leading domain-specific baseline ChatScene and a 43.0% reduction relative to the best-performing general-purpose LLM, Gemini 3.0 Pro. Concurrently, TextScene attains an ILD of 0.56, surpassing ChatScene by 24.4% and Text2Scenario by 33.3%, while remaining competitive with general-purpose LLMs that possess substantially larger parameter scales. This dual superiority in adversarial risk and scenario diversity confirms that integrating empirical accident cases with causal trigger logic effectively injects high-risk factors into the generated scenarios while simultaneously preserving behavioral diversity.

4.2.2. Assessment of Scenario Quality and Diversity

Beyond executability, the testing value—defined by risk and diversity—is critical for evaluating simulation frameworks. We quantitatively assessed scenarios generated from 20 functional tasks using TextScene, with results illustrated in Figure 9.

Risk Analysis: To evaluate scenario challenge, we classified outputs into low, medium, and high risk levels. The risk level is determined by a weighted combination of scores independently assigned by human experts and the LLM to each scenario description. Specifically, both the human experts and the LLM rate each scenario on a scale of 1 to 10,

and the final score is calculated using weights of 0.7 and 0.3, respectively. Scores of 1-5, 6-7, and 8-10 correspond to low-, medium-, and high-risk scenarios. To improve scoring accuracy, the LLM is additionally provided with calibrated reference examples representing low risk (score 1) and high risk (score 10), enabling more accurate and consistent risk classification.

As shown in Figure 9 (a), medium-to-high risk scenarios constitute over 50% across all tasks. Notably, in safety-critical tasks like "Emergency Hazard Avoidance" and "Response to Lead Vehicle Cut-out", high-risk scenarios (purple bars) exceed 50%. This distribution validates the efficacy of our accident scenario and trigger condition libraries. By injecting real-world accident priors and risk factors via few-shot prompting, the framework autonomously synthesizes high-value test sets rich in long-tail scenarios.

Table 4

Risk-stratified executability and fidelity of TextScene across 1,000 evaluation scenario.

Risk Level	Eval Pairs	CR (%)↑	SSR (%)↑	SC (1-5)↑
Low	100	95.0	94.0	4.68
Medium	100	92.0	90.0	4.54
High	100	88.0	84.0	4.46

To further evaluate framework performance across varying risk levels, we stratified executability and fidelity metrics into three risk categories, as presented in Table 4. As expected, performance degrades with increasing risk level. CR declines from 95.0% for low-risk scenarios to 88.0% for high-risk scenarios, while SSR decreases from 94.0% to 84.0%, reflecting the heightened interaction complexity and tighter spatiotemporal constraints inherent in high-risk test cases. SC exhibits a minimal decline from 4.68 to 4.46, indicating that TextScene maintains strong semantic fidelity even under challenging adversarial conditions. Notably, the high-risk SSR of 84.0% remains substantially higher than ChatScene's overall SSR of 61.3%, confirming that our framework sustains robust executability across the full spectrum of testing difficulty levels.

Diversity Analysis: Diversity is essential to prevent algorithmic overfitting. As shown in Figure 9 (b), ILD scores remain consistently high across all test tasks (*average* > 0.5), with long-horizon tasks like "Highway Off-Ramp Exiting" exhibiting exceptional diversity. This is attributed to the RAG mechanism, which retrieves distinct accident cases and injects varied trigger conditions. Synergized with the LLM's inherent reasoning divergence, this ensures that even identical functional requirements yield scenarios with unique spatiotemporal characteristics.

4.3. Ablation Study

To isolate the contributions of TextScene's core mechanisms, we designed five ablation variants by systematically removing or partially disabling the knowledge-retrieval, map-topology alignment, and instruction-tuning components:

- **Without Knowledge RAG:** This variant removes both accident-case retrieval and trigger-condition retrieval. By relying exclusively on the LLM's parametric knowledge for logical scenario generation, it evaluates the overall contribution of external knowledge grounding.
- **Accident-only RAG:** This variant retains only the accident-case retrieval stream while disabling trigger-condition retrieval. It assesses the contribution of empirical accident priors to semantic-behavioral fidelity and realistic interaction patterns.
- **Trigger-only RAG:** This variant retains only the trigger-condition retrieval stream while disabling accident-case retrieval. It evaluates the contribution of causal risk conditions to testing criticality and low-frequency scenario variation.
- **Without Map-Topology Abstraction:** This variant discards the Map-Topology Abstraction module. Instead, the LLM is provided with raw OpenDRIVE fragments or coarse-grained navigation instructions. This setup verifies the critical role of semantic mapping in improving physical plausibility.
- **Without Instruction Tuning:** This variant employs the untuned base model (Qwen3-32B) with a few-shot prompting strategy. This comparison validates the necessity of the fine-tuning mechanism for enforcing syntactic constraints.

Table 5

Ablation results for the core components of TextScene. **AC**: accident-case retrieval stream; **TC**: trigger-condition retrieval stream.

Method Variant	Components				Executability		Fidelity	Testing Value	
	SFT	Map	AC	TC	CR (%)↑	SSR (%)↑	SC (1-5)↑	m-TTC (s)↓	ILD↑
Without Instruction Tuning	-	✓	✓	✓	49.8 (↓40.8)	37.6 (↓50.3)	4.47 (↓0.04)	2.35 (↑0.04)	0.52 (↓0.04)
Without Map-Topology Abstraction	✓	-	✓	✓	90.3 (↓0.3)	26.5 (↓61.4)	4.43 (↓0.08)	2.45 (↑0.14)	0.52 (↓0.04)
Without Knowledge RAG	✓	✓	-	-	90.4 (↓0.2)	88.3 (↑0.4)	3.25 (↓1.26)	4.95 (↑2.64)	0.30 (↓0.26)
Accident-only RAG	✓	✓	✓	-	90.5 (↓0.1)	88.1 (↑0.2)	4.23 (↓0.28)	3.82 (↑1.51)	0.46 (↓0.10)
Trigger-only RAG	✓	✓	-	✓	90.4 (↓0.2)	87.6 (↓0.3)	3.88 (↓0.63)	2.55 (↑0.24)	0.51 (↓0.05)
TextScene (Full)	✓	✓	✓	✓	90.6	87.9	4.51	2.31	0.56

4.3.1. Ablation Analysis

The experimental results of the ablation study are shown in Table 5.

Impact of Knowledge RAG: Removing the RAG module yields a marginal SSR improvement of 0.4 percentage points. This ostensible gain is attributable to a shift toward spatially conservative, low-interaction scenarios that are inherently more amenable to simulation validation, rather than a genuine enhancement in generative capability. The concurrent degradation in m-TTC (+2.64s) and ILD (-0.26) confirms that this variant produces scenarios of greater compilability but substantially diminished adversarial challenge, thereby compromising the primary objective of safety-critical test generation. To isolate the contribution of each retrieval stream, we evaluated Accident-only and Trigger-only RAG configurations. Accident-only RAG raises SC to 4.23 but leaves m-TTC relatively high at 3.82s. Conversely, Trigger-only RAG lowers m-TTC to 2.55s and improves ILD to 0.51. The full dual-stream model achieves the optimal balance, confirming that the two knowledge sources are complementary: accident priors anchor semantic plausibility, while trigger logic enforces adversarial intensity. Thus, TextScene prioritizes high testing value scenarios through this heterogeneous dual-stream design, accepting a negligible SSR trade-off for superior semantic consistency and adversarial intensity.

Impact of Map-Topology Abstraction: Ablating Map-Topology Abstraction reveals that syntactic validity does not guarantee physical feasibility. While Instruction Tuning maintains a high CR of 90.3%, SSR decreases sharply from 61.4% to 26.5%. Lacking topological constraints, the model suffers from severe spatial hallucinations, generating syntactically correct but physically implausible code (e.g., entities in static infrastructure or off-road). This confirms Map-Topology Abstraction as the essential grounding mechanism anchoring logical scenarios to physical reality.

Impact of Instruction Tuning: Removing instruction tuning causes a collapse in syntactic validity. CR degrades by 40.8% to 49.8%, dragging SSR down to 37.6%. This confirms that general-purpose LLMs struggle with the hierarchical complexity and strict constraints of OpenSCENARIO XML. Without domain-specific alignment, the model exhibits structural hallucinations—fabricating tags or omitting parameters—rendering the code unparseable.

4.4. Failure analysis

We categorized all failures observed across the 1,000 scenario evaluations into three types: OpenSCENARIO Syntax Failures, Spatial Hallucination Failures, and Semantically Incorrect but Executable scenarios. As detailed in Table 6, each scenario (12.1%) exhibit at least one failure mode. Because a single scenario may involve multiple failure modes, the subtype counts are reported in a multi-label manner.

OpenSCENARIO Syntax Failures are the dominant hard-failure mode, covering 94 scenarios (9.4%, or 77.7% of all hard failures). Invalid or missing required attributes account for the majority (58 cases), suggesting that fine-grained parameter constraints in OpenSCENARIO code remain challenging for the instruction-tuned model to satisfy consistently. **Spatial Hallucination Failures** account for 45 scenarios (4.5%), where syntactically valid code fails during simulation due to trajectory discontinuities, out-of-bound waypoints, or conflicting participant positions. As shown in Table 7, removing the map-topology module causes severe degradation across all four spatial indicators, confirming that explicit topological encoding is essential for physical grounding. **Semantically Incorrect but Executable** scenarios are the least prevalent (11 cases, 1.1%), occurring when generated behaviors deviate from the intended adversarial objective despite successful execution, most commonly through misspecified risk-triggering behavior (6 cases).

Table 6

Failure mode analysis of TextScene on 1,000 scenario-map evaluation pairs

Failure Category	Failure Sub-type	Count	Proportion (%)
OpenSCENARIO Syntax Failure	Hierarchical structure violation	23	2.3
	Invalid or missing required attribute	58	5.8
	Incomplete or truncated code block	13	1.3
Spatial Hallucination Failure	Road / lane ID reference mismatch	12	1.2
	Trajectory discontinuity or out-of-bound violation	20	2.0
	Conflicting positions among traffic participants	13	1.3
Semantically Incorrect but Executable	Incorrect participant type assignment	3	0.3
	Misspecified risk-triggering behavior	6	0.6
	Erroneous ego-vehicle maneuver intent	2	0.2

Table 7

Spatial hallucination failure analysis over 1,000 evaluation scenarios. Lower is better for ORR, OLR, LVR, and VOR.

Configuration	ORR (↓)	OLR (↓)	LVR (↓)	VOR (↓)
without Map-Topology	10.2%	17.0%	39.6%	6.7%
TextScene (Full)	0.3%	0.9%	2.0%	1.3%
Improvement	-9.9%	-16.1%	-37.6%	-5.4%

To further validate the Map-Topology Abstraction module and characterize spatial hallucination failure modes, we conducted a fine-grained analysis across all 1,000 evaluation scenarios. We evaluated four spatial hallucination indicators—Off-Road Rate (ORR), Off-Lane Rate (OLR), Lane Violation Rate (LVR), and Vehicle Overlap Rate (VOR)—comparing the full TextScene framework against its ablated variant without map-topology alignment. Results are presented in Table 7.

These metrics quantify distinct aspects of physical plausibility. **ORR** measures the proportion of scenarios in which at least one participant is initialized at incorrect road IDs. **OLR** quantifies scenarios where entities are placed on nonexistent or incorrect lane IDs. **LVR** captures trajectory-level errors, such as lane changes crossing solid boundaries or transitions between non-adjacent lanes. **VOR** denotes initialization conflicts where two or more participants occupy overlapping positions, causing immediate collision.

As shown in Table 7, removing map-topology alignment causes severe degradation across all spatial dimensions: ORR rises from 0.3% to 10.2%, OLR from 0.9% to 17.0%, LVR from 2.0% to 39.6%, and VOR from 1.3% to 6.7%. LVR exhibits the most pronounced degradation—nearly 40% of scenarios produce invalid trajectories in the absence of topological constraints. These results confirm that explicit topological encoding is essential for geometric coherence: without it, the LLM fails to maintain spatial consistency, producing widespread physical violations despite syntactically well-formed outputs.

4.5. Framework Generalizability Analysis

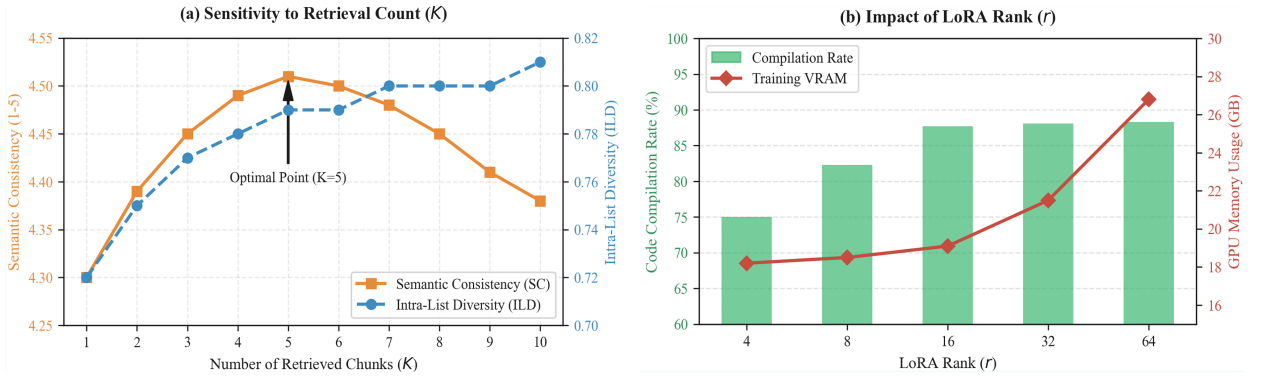
To verify the model-agnostic nature of TextScene, we applied our full methodology (RAG, Map-Topology Abstraction, Instruction Tuning) to Llama, Mistral, and Gemma architectures. Results are summarized in Table 8.

Experiments demonstrate the robust generality of TextScene across diverse architectures and scales, as shown in Table 8). Notably, TextScene empowers lightweight models to surpass unoptimized larger baselines; for instance, Llama-3.1-8B with TextScene outperforms the Llama-3.1-70B baseline by 13.3 percentage points in CR and 6.9 percentage points in SSR. Furthermore, the framework enables exceptional efficiency, with the medium-sized Qwen3-32B (SSR 87.9%) exceeding both Llama-3.1-70B (83.1%) and Mistral-8x7B (82.4%). For Qwen3-32B, TextScene reduces m-TTC by 1.94s and improves ILD by 0.22, further confirming enhanced adversarial criticality and scenario diversity. These results indicate that TextScene enables lightweight models to approach the generation quality of large-scale counterparts: Qwen3-32B with TextScene achieves an SSR of 87.9%, surpassing the Llama-3.1-70B baseline and Mistral-8x7B while requiring fewer than half the parameters. This parameter efficiency reduces GPU memory requirements during inference, lowering the hardware threshold for industrial deployment.

Table 8

Generalizability analysis across different model architecture and parameter scales.

Backbone	Params	Method	Executability		Fidelity	Testing Value	
			CR (%)↑	SSR (%)↑	SC (1-5)↑	m-TTC (s)↓	ILD↑
Llama-3.1-8B (Grattafiori et al., 2024)	8B	Few-shot Baseline + TextScene	25.4 68.5	12.1 45.4	3.42 3.95	5.85 2.85	0.29 0.45
Llama-3.1-70B (Grattafiori et al., 2024)	70B	Few-shot Baseline + TextScene	55.2 89.2	38.5 83.1	4.25 4.48	4.52 2.38	0.46 0.54
Mistral-v0.3-7B (Jiang et al., 2024)	7B	Few-shot Baseline + TextScene	28.1 74.5	14.2 42.5	3.55 4.02	5.62 2.76	0.32 0.48
Mistral-8x7B (Jiang et al., 2024)	47B	Few-shot Baseline + TextScene	52.8 88.5	36.5 82.4	4.18 4.42	4.68 2.41	0.44 0.51
Gemma-2-9B (Team et al., 2025)	9B	Few-shot Baseline + TextScene	22.8 72.4	10.5 48.8	3.38 3.88	6.10 2.92	0.25 0.42
Gemma-2-27B (Team et al., 2025)	27B	Few-shot Baseline + TextScene	45.6 86.8	30.4 79.5	4.10 4.35	4.95 2.45	0.42 0.52
Qwen3.5-7B (Yang et al., 2025)	7B	Few-shot Baseline + TextScene	31.5 81.5	18.2 58.5	3.62 4.15	5.45 2.68	0.35 0.50
Qwen3-32B (Yang et al., 2025)	32B	Few-shot Baseline + TextScene (Ours)	58.5 90.6	42.6 87.9	4.30 4.51	4.25 2.31	0.34 0.56

**Fig. 10.** Hyperparameter Sensitivity Analysis. Subfigure (a) illustrates the impact of retrieval count K ; Subfigure (b) illustrates the impact of LoRA rank r .

4.6. Parameter Sensitivity Analysis

To identify optimal configuration trade-offs, we conducted a sensitivity analysis on two critical hyperparameters: the Retrieval Count K for RAG and the LoRA Rank r . The results are presented in Figure 10.

4.6.1. Sensitivity to Retrieval Count (K)

The analysis of retrieval count $K \in [1, 10]$ reveals a critical trade-off between context grounding and noise introduction. As shown in Figure 10 (a), Semantic Consistency (SC) improves initially, peaking at 4.51 with $K = 5$, suggesting that adequate real-world references effectively anchor the generation logic; however, increasing K beyond this threshold causes SC to decline to 4.38, as excessive context introduces irrelevant noise. Simultaneously, Intra-List Diversity (ILD) exhibits a sharp initial rise from 0.72 to 0.79 at $K = 5$. However, for $K > 5$, this growth saturates, yielding negligible marginal gains. Consequently, we adopt $K = 5$ as the optimal trade-off: it achieves the highest SC while maintaining a high ILD, and avoids the additional computational overhead and contextual noise introduced by larger retrieval counts.

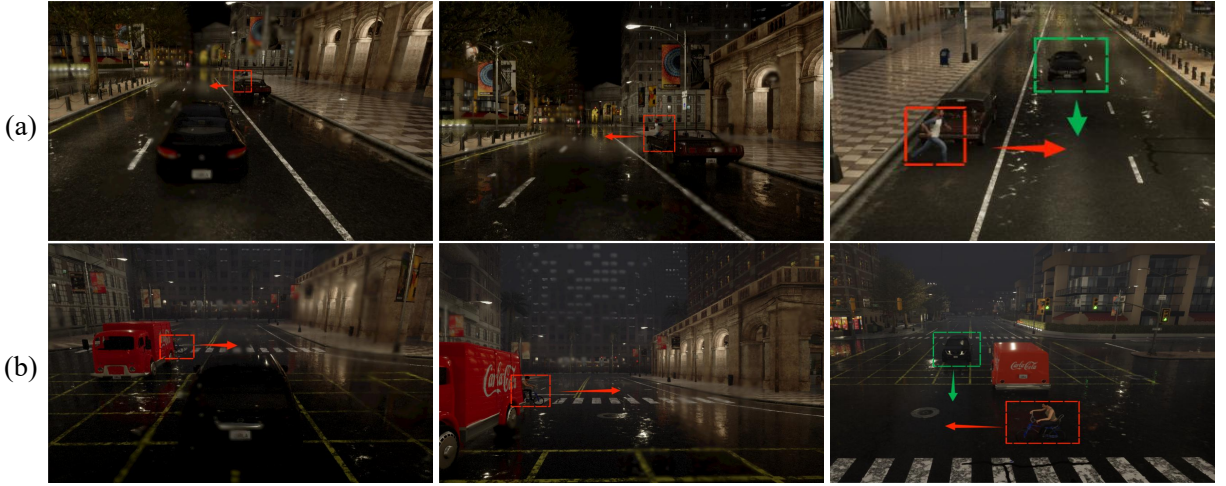


Fig. 11. Qualitative Visualization of Complex Safety-Critical Scenarios Generated by TextScene. Row (a): The "Ghost Probe" scenario, where a pedestrian suddenly darts out from the blind spot of a roadside parked vehicle in a rainy night setting. **Row (b):** An "Occluded Crossing" scenario, involving a cyclist crossing blindly from behind a large truck.

4.6.2. Impact of LoRA Rank (r)

We further evaluated the impact of LoRA rank r on syntactic compliance and resource efficiency. As shown in Figure 10 (b), the Compilation Rate (CR) increases from 75.0% at $r = 4$ to 87.6% at $r = 16$, indicating that a minimal parameter threshold is essential for the model to capture the intricate nesting structures of OpenSCENARIO. However, increasing r beyond 16 yields negligible performance gains (plateauing at 88.3%) while incurring a disproportionate memory penalty, with training VRAM escalating from 19.1 GB to 26.8 GB. Therefore, $r = 16$ is selected as the optimal configuration, ensuring robust code generation capabilities while preserving the lightweight, consumer-grade hardware compatibility of the fine-tuning process.

4.7. Visualization and Case Studies

To visually substantiate the capability of the TextScene framework in translating natural language functional requirements into high-fidelity physical simulations, we present visualization results within the CARLA simulator. As illustrated in Figure 11, we conducted a detailed case study based on the user instruction "generate a scenario for testing Automatic Emergency Braking (AEB)".

Upon receiving the AEB test requirements, the RAG module retrieves relevant risk patterns and potential hazards from the real-world accident repository and trigger condition database. Guided by this retrieval-augmented context, the LLM synthesizes multiple critical scenarios tailored for AEB validation. Figure 11 depicts the generated high-risk "ghost probe" interactions (i.e., sudden pedestrian emergence from occlusion). In Figure 11 (a), the system automatically positions a stationary vehicle (indicated by the red bounding box) as a static occlusion source. The LLM precisely calculates the Time-to-Collision (TTC) between the ego vehicle and the Vulnerable Road User (VRU), orchestrating the pedestrian to emerge abruptly exactly as the ego vehicle approaches the blind spot (marked by the red trajectory). This precise temporal synchronization demonstrates the model's robust capacity to manage dynamic spatiotemporal risks. Furthermore, Figure 11 (b) presents a variant of this scenario involving dynamic occlusion caused by a large moving vehicle.

5. Conclusion

This paper proposes **TextScene**, a knowledge-enhanced framework designed to bridge the semantic-to-physical gap inherent in translating natural language into autonomous driving scenarios.

Through **Real-Data-Grounded Dual-Stream Retrieval-Augmented Scene Parsing**, we systematically incorporate real-world accident statistics and causal risk mechanisms into the generation pipeline, effectively mitigating the hallucinations and triviality often exhibited by general-purpose LLMs. To address spatial inconsistencies, our proposed **Map-Topology Abstraction Algorithm** transforms intricate geometric calculations into explicit topological reasoning, thereby markedly reducing physical anomalies such as vehicle overlapping and retrograde driving.

Furthermore, through **Syntax-Constrained Instruction-Tuning Paradigm**, we establish a reliable mapping from physical parameters to the rigorous OpenSCENARIO XML syntax, achieving a Compilation Rate of **90.6%** and a Simulation Success Rate of **87.9%**. Extensive empirical evaluations demonstrate that TextScene achieves a CR of 90.6% and SSR of 87.9%, outperforming all evaluated general-purpose LLMs and domain-specific baselines in executability. Furthermore, Qwen3-32B with TextScene surpasses the Llama-3.1-70B baseline by 4.8 percentage points in SSR, suggesting that structured domain knowledge injection can partially compensate for the capacity advantage of larger parameter-scale models in this task.

In future work, we plan to integrate simulation execution feedback into a Reinforcement Learning from Human Feedback (RLHF) loop, facilitating the automated, continuous iteration and adversarial evolution of generated scenarios.

CRediT authorship contribution statement

Yuhao Jiang: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Qiu Chuang:** Writing – review & editing, Validation. **Yang Chen:** Writing – review & editing, Methodology, Validation. **Kai Chen:** Writing – review & editing, Conceptualization, Visualization. **Ke Wang:** Writing – review & editing, Project administration, Resources, Funding acquisition.

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